

in the deceased cohort likely stems from acidosis secondary to acute respiratory distress syndrome, multi-organ failure, DIC, or other pre-mortality syndromes. When trained on blood laboratory values, M3MIL proves adept at predicting patient mortality with high accuracy while highlighting the underlying morphologies driving changes in patient condition; image-based analysis and pathophysiological predictions set it apart from other clinical tools for predicting mortality.

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3 Identifying Hospitalized Patients with Acute Decompensation Risk from Peripheral Blood Smears and Laboratory Results using Multimodal, Multienncoder, Multiple Instance Learning (M3MIL)

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For hospitalized patients in critical condition, anticipating clinical deterioration and identifying its cause is imperative for timely interventions, such as escalation of care, ventilatory support, or sepsis protocol; we therefore propose a deep learning approach that uses peripheral blood smear (PBS) images, complete blood counts (CBCs), and blood chemistry panels to quantify a patient's mortality risk and identify underlying syndromes through analysis of cellular morphology, thereby informing clinical decisions that improve patient outcomes. Using a database of digitized peripheral blood smears imaged by CellaVision and lab reports for a cohort of deceased and control patients randomly sampled from an inpatient population, we trained a novel multimodal, multienncoder, multiple instance learning model (M3MIL) to predict the probability of hospital mortality. On a held-out test set, the model achieves AUC = 0.98, sensitivity = 0.88, and specificity = 0.98. Furthermore, the self-attention mechanism captures the relative importance of predictors, enabling the identification of mortality syndromes through morphological frequencies. When comparing the likelihood of cell morphologies in aggregate between the deceased and control cohort, we observed a higher relative attention on schistocytes and thrombocyte aggregations (suggesting DIC or MAHA), band neutrophils (bacterial sepsis), variant lymphocytes (viral sepsis), and echinocytes. While echinocytes are associated with chronic kidney and liver failure, we found the high echinocyte signal